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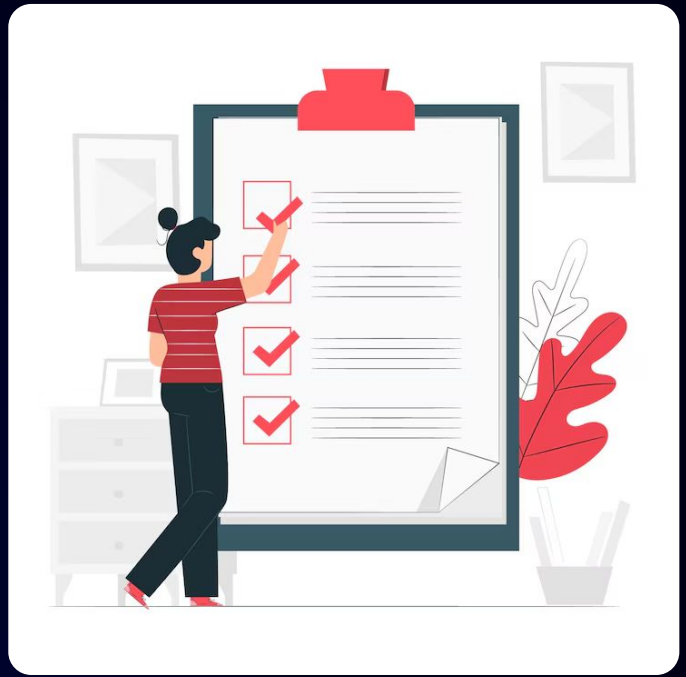
DOM Manipulation

Learning Objectives

- Visualize and understand the structure and working mechanism of the DOM.
- Learn essential DOM methods for selecting and accessing HTML elements.
- Explore internal HTML object collections and their role in the DOM.
- Manipulate content and attributes of DOM elements using intermediate DOM methods.
- Create and manipulate DOM elements programmatically
- Apply advanced styling techniques and manage classes using the DOM.

Prerequisites

- Basic understanding of HTML elements, tags, and page structure.
- Fundamental knowledge of CSS selectors, classes, and styling concepts.
- Familiarity with JavaScript basics such as variables, functions, loops, and conditions.
- Understanding of JavaScript events like click and input events.
- Basic knowledge of browser developer tools and webpage inspection.



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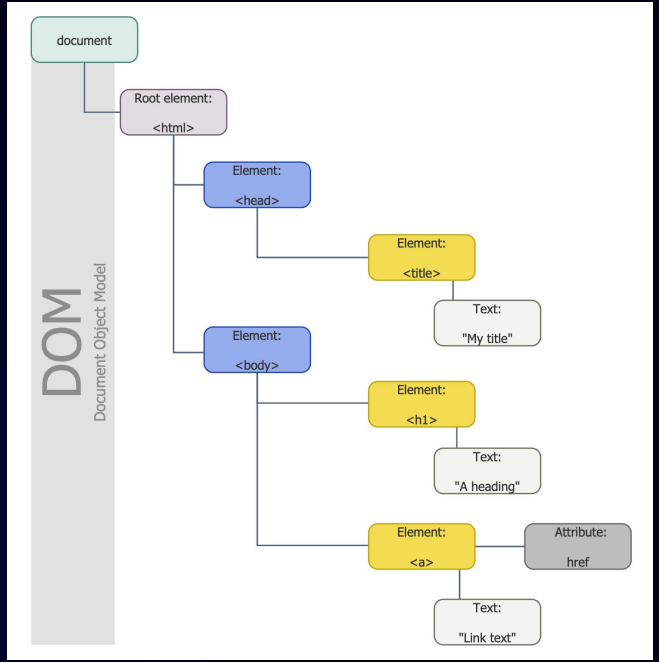
Visualization of the DOM and Its Working

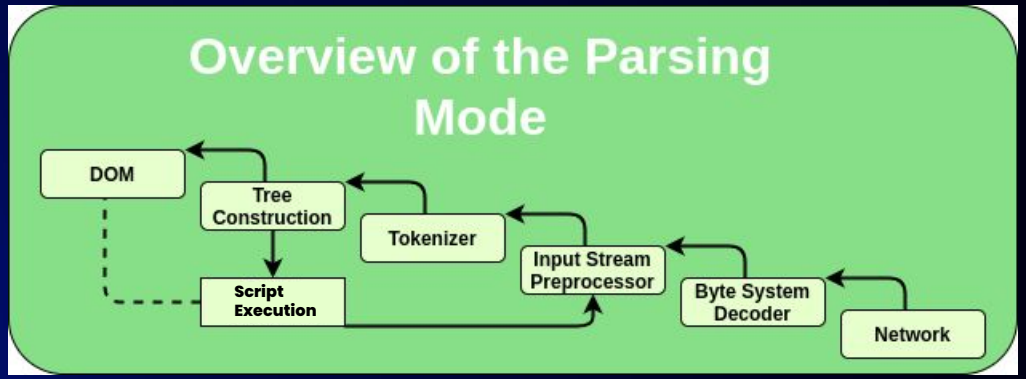
DOM (Document Object Model)

- **Definition**: The Document Object Model (DOM) is an API for HTML and XML documents. It provides a structured representation of a document as a tree of objects.
- **Purpose**: It allows programming languages to interact with the document structure, style, and content.
- **Components**:
  - **Nodes**: The building blocks of the DOM. Includes elements, attributes, and text.
  - **Tree Structure**: Represents HTML as a hierarchical tree where each node is an object.



How the DOM Works

- HTML Parsing: Browsers read HTML markup and convert it into a structured format.
- DOM Construction: The browser constructs the DOM tree by creating nodes for elements, attributes, and text.
- Rendering: The DOM tree is then used to render the visual representation of the webpage.
- Updates: Dynamic changes in the DOM reflect immediately on the webpage.



Pop Quiz

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Q.What does DOM stand for?

A

Document Object Model

B

Digital Object Model

Pop Quiz

PW SKILLS

Q.What does DOM stand for?

A

Document Object Model

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Digital Object Model

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DOM Basic Methods

Essential DOM Methods

- **getElementById()**
  - Purpose: Selects a single element with the specified ID.
  - Syntax: document.getElementById('elementId')
  - Example: let myElement = document.getElementById('header')

- **getElementsByClassName()**
  - Purpose: Selects all elements with the specified class name.
  - Syntax: document.getElementsByClassName('className')
  - Example: let items = document.getElementsByClassName('menu-item')

Essential DOM Methods

- **getElementsByTagName()**
  - Purpose: Selects all elements with the specified tag name.
  - Syntax: document.getElementsByTagName('tagName')
  - Example: let paragraphs = document.getElementsByTagName('p')

- **querySelector()**
  - Purpose: Selects the first element that matches a specified CSS selector.
  - Syntax: document.querySelector('selector')
  - Example: let firstItem = document.querySelector('.list-item')

Essential DOM Methods

- **querySelectorAll()**
  - Purpose: Selects all elements that match a specified CSS selector.
  - Syntax: document.querySelectorAll('selector')
  - Example: let allItems = document.querySelectorAll('li')

Hands-On | Working with DOM Selectors

Pop Quiz

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Q.Which method returns a NodeList of all elements matching a CSS selector?

A

getElementById()

B

querySelectorAll()

Pop Quiz

PW SKILLS

Q.Which method returns a NodeList of all elements matching a CSS selector?

A

getElementById()

B

querySelectorAll()

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Understanding Internal HTML Object Collections

HTML Object Collections

- **HTMLCollection:**
  - Definition: A live collection of HTML elements.
  - Characteristics: Automatically updates when the document changes.
  - Usage: Often returned by methods like getElementsByClassName() and getElementsByTagName().

- **NodeList:**
  - Definition: A collection of nodes (could be elements or other node types).
  - Characteristics: Can be live or static. Static NodeList doesn't update with document changes.
  - Usage: Returned by methods like querySelectorAll() and childNodes.

HTML Object Collections

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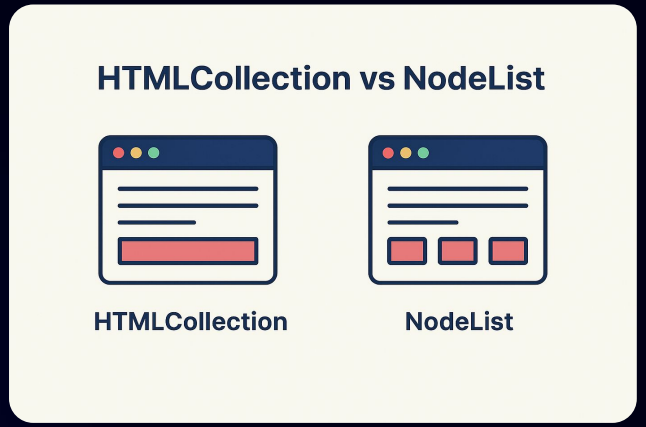
HTMLCollection	NodeList
Contains only HTML elements	Can contain elements, text nodes, comments
Live collection	Can be live or static
Returned by methods like `getElementsByClassName()`	Returned by `querySelectorAll()`



HTML Object Collections example

```
<button class="btn">First button</button>
<button class="btn">Second button</button>
<button class="btn">Third button</button>
```

In this example, we have three button elements. Each has a class of btn. Now, let's select the buttons using the getElementsByClassName method.

```
const buttonElements = document.getElementsByClassName('btn')
console.log(buttonElements)
```

```
▶ HTMLCollection(3) [button.btn, button.btn, button.btn]
```

Hands-On | Understanding Internal HTML Object Collections

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Q.Is an HTMLCollection live or static?

A

live collection

B

Static collection

Pop Quiz

PW SKILLS

Q.Is an HTMLCollection live or static?

A

live collection

B

Static collection

HTMLCollection is a **live collection**, meaning it automatically updates when elements are added or removed from the DOM.

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Intermediate DOM Methods

innerText

- Sets or retrieves the text content of an element, excluding HTML tags.
- **Usage**

```
element.innerText = "New text content";
var text = element.innerText;
```

innerHTML

- Sets or retrieves the HTML content of an element, including HTML tags.
- **Usage**

```
element.innerHTML = "<b>Bold text</b>";
var html = element.innerHTML;
```

textContent

- Sets or retrieves the text content of an element, including hidden elements.
- **Usage**

```
element.textContent = "New text content";
var text = element.textContent;
```

setAttribute()

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- Sets the value of an attribute on an element.
- **Usage**

```
element.setAttribute("data-custom", "value");
```

getAttribute()

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- Retrieves the value of an attribute from an element.
- **Usage**

```
var value = element.getAttribute("data-custom");
```

Hands-On | Intermediate DOM Methods

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Q. What happens if the attribute already exists?

A

It updates (overwrites) the existing value.

B

The existing value does not get overwritten.

Pop Quiz

PW SKILLS

Q. What happens if the attribute already exists?

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B

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Take A 5-Minute Break!

- Stretch and relax
- Hydrate
- Clear your mind
- Be back in 5 minutes

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Element Creation and Manipulation

Creating and Manipulating DOM Elements

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- **createElement()**
  - Purpose: Creates a new HTML element.
  - Syntax: document.createElement('tagName')
  - Example: let newDiv = document.createElement('div')

- **appendChild()**
  - Purpose: Adds a new child node to a parent node.
  - Syntax: parentNode.appendChild(childNode)
  - Example: document.body.appendChild(newDiv)

```
// Create element:
const para = document.createElement("p");
para.innerText = "This is a paragraph.";

// Append to body:
document.body.appendChild(para);
```

Creating and Manipulating DOM Elements

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- **append()**
  - Purpose: Adds new nodes or text to the end of a parent node.
  - Syntax: parentNode.append(node1, node2, ...)
  - Example: parent.append('New text', newElement)

```
let div = document.createElement("div");
let p = document.createElement("p");
div.append("Some text", p);
```

- **removeChild()**
  - Purpose: Removes a child node from a parent node.
  - Syntax: parentNode.removeChild(childNode)
  - Example: parent.removeChild(childToRemove)

```
const list = document.getElementById("myList");
list.removeChild(list.firstElementChild);
```

Creating and Manipulating DOM Elements

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- **replaceChild()**
  - Purpose: Replaces one child node with another.
  - Syntax: parentNode.replaceChild(newChild, oldChild)
  - Example: parent.replaceChild(newElement, oldElement)

```
// Select first child element:
const element = document.getElementById("myList").children[0];

// Create a new text node:
const newNode = document.createTextNode("Water");

// Replace the text node:
element.replaceChild(newNode, element.childNodes[0]);
```

Hands-On | Element Creation and Manipulation

Pop Quiz

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Q.What is the correct order of DOM element creation and insertion?

A

1. Create element → createElement()
2. Add content/attributes → textContent, setAttribute()
3. Attach to DOM → appendChild() / append()

B

1. Create element → createElement()
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Pop Quiz

PW SKILLS

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3. Add content/attributes → textContent, setAttribute()

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DOM Styling and Classes

style Property

- Directly sets inline styles on an element.
- **Usage**

```
element.style.color = "red";
```

classList.add()

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- **Definition:** Adds one or more classes to an element.

- **Usage**

```
element.classList.add("new-class");
```

classList.remove()

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- **Definition:** Removes one or more classes from an element.

- **Usage**

```
element.classList.remove("old-class");
```

classList.toggle()

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- **Definition:** Toggles a class on or off for an element.

- **Usage**

```
element.classList.toggle("active");
```

classList.contains()

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- **Definition:** Checks if an element contains a specific class.

- **Usage**

```
var hasClass = element.classList.contains("active");
```

Practical Examples

- **Creating a Class**

- **CSS**

```
<div id="box" class="box"></div>
```

- **JavaScript**

```
var box = document.getElementById("box");
box.classList.add("highlight");
```

Pop Quiz

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Q. What method is used to change the src attribute of an image element?

A

innerHTML

B

setAttribute()

Pop Quiz

PW SKILLS

Q. What method is used to change the src attribute of an image element?

A

innerHTML

B

setAttribute()

PW SKILLS

Guided Practice Tasks on **DOM Manipulation Deep Dive**

Summary

In this lecture, we have covered:

- The DOM represents an HTML page as a tree structure that JavaScript can access and modify.
- Elements are selected using getElementById, querySelector, and querySelectorAll.
- Content can be changed using innerText, textContent, and innerHTML.
- Attributes are handled using setAttribute() and getAttribute().
- New elements are created with createElement() and added/removed using append() and removeChild().
- CSS classes are managed using classList, and styles are changed using the style property.

DOUBT RESOLUTION

Important

- Complete the post-class assessment
- Complete assignments (if any)
- Practice the concepts and techniques taught in this session
- Review your lecture notes
- Note down questions and queries regarding this session and consult the teaching assistants.

THANK YOU